

For my term project, I will investigate the quantum tunneling behavior in a transition metal complex, specifically $[\text{Fe}(\text{CO})_5]$, which has been studied for its tunneling properties in various reaction mechanisms. This complex is known for its ability to undergo a variety of reactions via quantum tunneling, making it an ideal candidate for computational study. Using Gaussian software, I will apply both HF and DFT calculations to explore the energy barriers associated with tunneling and to understand the factors influencing this phenomenon.

The computational approach will begin with geometry optimization of the $[\text{Fe}(\text{CO})_5]$ complex using both the HF and DFT methods. For the DFT calculations, I will employ the B3LYP functional, which is widely used for modeling transition metal complexes due to its accurate treatment of electron correlation and exchange interactions. The 6-31G(d) basis set will be used for the lighter atoms, and a triple-zeta basis set (def2-TZVP) will be applied to the iron atom to better capture the subtleties of its electronic structure. Hartree-Fock calculations will be performed using the same basis set to compare the results and assess the performance of each method in describing tunneling behavior.

Once the optimized structures are obtained, I will calculate the potential energy surface (PES) for the complex, particularly focusing on regions where quantum tunneling may occur. This will involve scanning along the reaction coordinate, which is typically the displacement of the CO ligands, and calculating the energies at various points along the path. The resulting PES will allow for the identification of potential barriers that could facilitate or inhibit tunneling. To analyze tunneling, I will use the Wentzel–Kramers–Brillouin (WKB) approximation, which is commonly applied in quantum chemistry to estimate tunneling probabilities through potential barriers. This approach is particularly suitable for systems like transition metal complexes, where tunneling is expected to play a role in reaction mechanisms, such as in redox chemistry and catalysis.

I want use the both HF and DFT levels of theory to get insight into the accuracy and computational cost of each method but not sure if that is a wise decision (from what I gather DFT is generally more efficient and can offer better results for electron correlation but HF may be less computationally expensive but could miss some important electron interactions that affect tunneling). The findings will be validated against available experimental data for similar transition metal systems to ensure the robustness of the computational models.

By the end of this project, I expect to provide a comprehensive analysis of the quantum tunneling behavior in the $[\text{Fe}(\text{CO})_5]$ complex, comparing the results from both HF and DFT methods. This study will help elucidate the role of electronic structure and reaction coordinate in determining tunneling rates and energy barriers. Additionally, this work will contribute to the broader understanding of tunneling in transition metal chemistry and may have implications for the design of catalysts or materials where tunneling is a critical factor.

The theoretical methods I will use are well-established in the literature. The B3LYP functional and the 6-31G(d) basis set are commonly applied to transition metal complexes and have been shown to provide a reliable description of the electronic structure and reaction dynamics in these systems (Becke, 1993; Lee et al., 1988). Hartree-Fock methods have also been used extensively in computational studies of tunneling in chemical reactions, particularly as a benchmark for more computationally expensive methods (Szabo & Ostlund, 1996).

References:

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- Szabo, A., & Ostlund, N. S. (1996). *Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory*. Dover Publications.